

Antifungal agents

Drugs/drug class: Amphotericin B, Flucytosine, Azoles, Echinocandins

Amphotericin B

- Amphotericin B is produced by *Streptomyces nodosus*.
- It is nearly insoluble in water
 - Prepared as a **colloidal suspension** of amphotericin B and
 - sodium deoxycholate** for IV.

Lipid Formulation of Amphotericin B

- ❖ Liposomal amphotericin preparations package the active drug in lipid delivery vehicles
- ❖ Formulated to **reduce drug-induced** renal impairment
 - lipid-packaged drug **binds to the mammalian** membrane less readily
 - Amphotericin binds to the lipids in vehicles with an affinity between that for fungal ergosterol and that for human cholesterol.
 - The lipid vehicle then serves as an amphotericin **reservoir**,
 - some fungi contain lipases that may liberate free amphotericin B directly at the site of infection.

Lipid Formulation of Amphotericin B

Lipid formulations are with preserved efficacy

- ✓ Less related toxicities
- ✓ Expensive
 - Limited to patients not responding/ tolerated conventional preparation

Amphotericin B

- ❖ Amphotericin B is poorly **absorbed from** the GIT
- ✓ Oral amphotericin B has effective only on fungi within the **lumen of GI**
- ✓ cannot be used for treatment of **systemic disease**.

PK

- ✓ IV..... 0.6 mg/kg/dresult in 0.3–1 mcg/mL
- ✓ Protein binding.....> 90 %
- ✓ The serum half-life > 15 days.
- ✓ widely distributed in most tissues, but
- ✓ only 2–3% reached in cerebrospinal fluid,
 - occasionally necessitating intrathecal therapy for certain types of fungal meningitis

Mechanisms of Action & Resistance

Amphotericin B is selective in its fungicidal effect

✓ it exploits the difference in lipid composition of fungal and mammalian cell membranes.

–Act on Ergosterol...a cell membrane sterol, is found in the cell membrane of fungi,

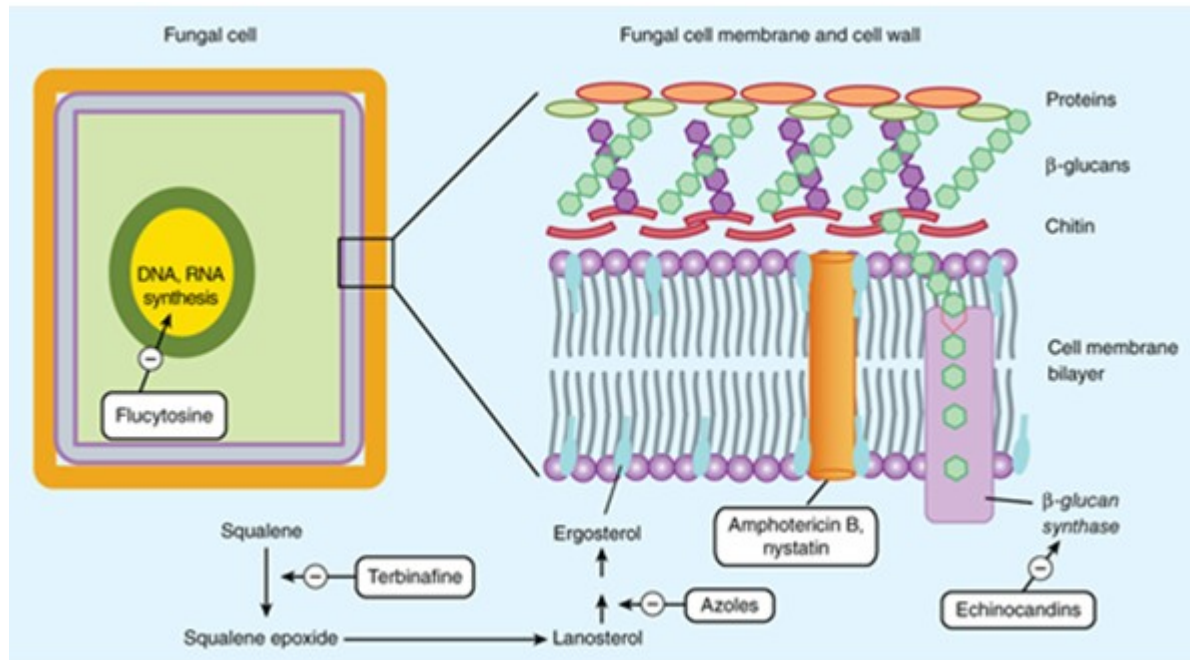
MOA....It binds to ergosterol and alters the permeability of the cell

Interaction/binding....

–with lipids (ergosterol) along the double bond–rich side

–with water molecules along the hydroxyl-rich side.

•This amphipathic characteristic facilitates pore ...leakage ofintracellular ions and macromolecules.....cell death.



Targets of antifungal drugs.

Mechanisms of Action & Resistance...

- ❖ Resistance to amphotericin B occurs
 - If ergosterol binding is impaired, due to
 - decreasing concentration of ergosterol/or modifications of sterol target

Antifungal Activity & Clinical Uses

- ❖ It has broadest spectrum and active against
 - **Yeasts**
 - *Candida albicans* and
 - *Cryptococcus neoformans*;
 - **the organisms causing endemic mycoses,**
 - *Histoplasma capsulatum*,
 - *Blastomyces dermatitidis*, and
 - *Coccidioides immitis*;
 - **pathogenic molds....***Aspergillus fumigatus* and the agents of mucormycosis.
- ❖ Resistance to *Candida lusitanae* and *Pseudallescheria boydii*

Clinical use

- ❖ Due to broad spectrum of activity ..used for all life-threatening mycotic infections,
 - but less toxic agents have largely replaced it for most conditions.
- ❖ It is often used as the initial induction regimen....then replaced by azoles (maintenance therapy)

Dose.... (0.5–1 mg/kg)...slow Iv infusion

Intrathecal amphotericin B.....fungal meningitisif not responded by other agents

Clinical use...

Local or topical amphotericin B

- ✓ For Mycotic corneal **ulcers** and **keratitis** can be cured with
 - direct subconjunctival injection.
- ✓ Fungal arthritis.....direct injection to joint.
- ✓ Candiduria.....bladder irrigation with amphotericin B

Adverse Effects

- ❖ The toxicity of amphotericin B can be
 - Infusion-related reactions
 - Cumulative toxicity

Infusion-related toxicity

Infusion-related reactions are nearly universal

- These are fever, chills, muscle spasms, vomiting, headache, and hypotension.
- ameliorated by slowing the infusion rate or decreasing the daily dose.
- Premedication with antipyretics, antihistamines, meperidine, or corticosteroids can be helpful. When
- starting therapy.....IV 1 mg (as a guide /know severity)

Cumulative toxicity

Renal damage is the most significant toxic reaction
reversible component

- decreased **renal perfusion** and represents a form of **prerenal renal failure**
- Can be attenuated with **sodium loading**,

irreversible component with (>4 g cumulative dose)

Adverse effects...

Anemia due to reduced erythropoietin production by damaged,
seizures

Flucytosine

Mechanism of action

- First it undergoes deamination (Flucytosine....to 5-FU)
- Fluorouracil....then.....metabolized to 5-fluorouracil-ribose monophosphate (5-FUMP) (by uracil phosphoribosyl transferase (UPRTase).
- 5-FUMP then is either incorporated into RNA (via synthesis of 5-fluorouridine triphosphate) or metabolized to 5-fluoro-2'-deoxyuridine-5'-monophosphate (5-FdUMP),
 - a potent inhibitor of thymidylate synthetase and thus of DNA synthesis.
 - The selective action of flucytosine is due to the lack of cytosine deaminase in mammalian cells.

Flucytosine

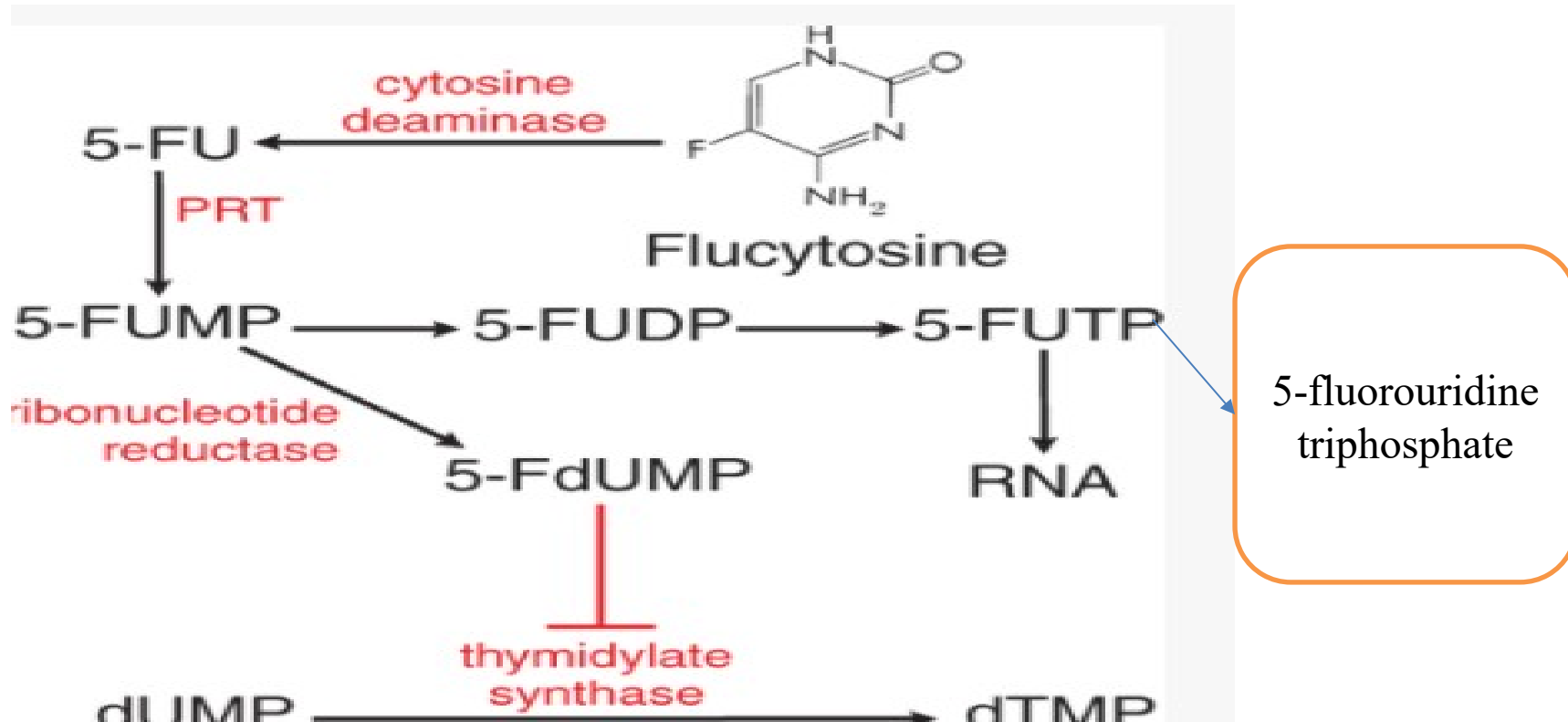


Figure: Mechanisms of Flucytosine

Chemistry & Pharmacokinetics

- ❖ **Dosage.....** 100 mg/kg/d in divided doses (normal renal function)
- ❖ **absorption**>90%,
- ❖ **half-life.....** 3–4 hours
- ❖ Toxicity is more likely to occur in AIDS patients and those with renal insufficiency concentration be..(50 and 100 mcg/m)L.

Mechanisms of resistance

- Resistance is thought to be mediated through altered metabolism of flucytosine,
 - develops rapidly in the course of flucytosine monotherapy.

Clinical Uses & Adverse Effects

❖ Spectrum of activity

- C neoformans,
- some Candida sp, and the
- dematiaceous molds that cause chromoblastomycosis.

❖ Flucytosine is rarely used as a single agent

- with amphotericin B.....for cryptococcal meningitis
- with Itraconazole....for chromoblastomycosis.

Adverse Effects

❖ ADR of flucytosine is resulted from Metabolism to.....fluorouracil. And the common adverse effects are

- Bone marrow toxicity with anemia,
 - leukopenia, and
 - thrombocytopenia.
- A form of toxic enterocolitis can occur.

Chemistry & Pharmacokinetics

can be classified as either

❖ imidazoles

- ✓ ketoconazole, miconazole, and clotrimazole
- ✓ Less selective to fungal enzyme
 - Higher incidence of drug interaction

❖ triazoles

itraconazole, fluconazole, voriconazole, isavuconazole, and posaconazole.

Mechanisms of Action & Resistance

MOA.... results from the reduction of ergosterol synthesis by inhibition of fungal cytochrome P450 enzymes

Selectivity..... their greater affinity for fungal than for human cytochrome P450 enzymes

Clinical Uses,

The spectrum of action.....They do have broad spectrum of activity

- ✓ Candida (*C neoformans*)
- ✓ mycoses (blastomycosis, coccidioidomycosis, histoplasmosis), the
- ✓ Dermatophytes
- ✓ Aspergillus..... itraconazole, posaconazole, isavuconazole, and voriconazole
- ✓ amphotericin-resistant organisms such as *P boydii*.

Adverse Effects, & Drug Interactions

- ❖ As a group....they are relatively nontoxic.

 - minor GI upset.....most common

- ✓ cause abnormalities in liver enzymes

- ✓ Hepatitis (very rarely)

- ❖ They are prone to drug interactions

- ❖ affect the mammalian P450 to some extent.

Ketoconazole

- ❖ the first oral azole introduced into clinical use.
- ❖ Compared to other triazoles by its
 - Greatly inhibit mammalian cytochrome P450 enzymes
 - less selective for fungal P450 than are the newer azoles.
 - It is no longer recommended for the treatment of **fungal nail or skin infections.**

Itraconazole

- ❖ available in oral and IV formulations and is used at a
—dosage of 100–400 mg/d.
- ❖ Drug interaction.....bioavailability reduced with
—rifamycins (rifampin, rifabutin, rifapentine).
- ❖ Bioavailability can be increased with **cyclodextrin** as
a carrier molecule

Itraconazole is the azole of choice for treatment of due to the

- | | |
|------------------------------|--------------------|
| —dimorphic fungi Histoplasma | —— Blastomyces |
| —Sporothrix | —— dermatophytoses |
| —Aspergillus sp | —— onychomycosis. |

•But replaced by voriconazol

Fluconazole

- ❖ available in **oral** and IV formulations
dosage of 100–800 mg/d.
- ❖ Oral bioavailability >> than itra and keto
- ❖ Drug interactions fluconazole has **the least effect** of all the azoles on hepatic microsomal enzymes.
- ❖ better gastrointestinal tolerance,
 - has the widest therapeutic index of the azoles,
 - more aggressive dosing in a variety of fungal infections possible.

Fluconazole...

Clinical use

- ❖ For secondary prophylaxis of cryptococcal meningitis.
- ❖ IV fluconazole = amphotericin B in treatment of candidemia in ICU, but
 - echinocandins may have superior activity for this indication.
- ❖ mucocutaneous candidiasis. But
 - no activity against *Aspergillus* or other filamentous fungi.
- ❖ Prophylactic....in bone marrow transplant recipients and AIDS patients

Voriconazole

- ✓ available in intravenous and oral (BA>90 %).
- ✓ dosage is 400 mg/d.
- ✓ Drug interaction.....It inhibits CYP3A4
 - (+) concentrations of cyclosporine, tacrolimus, and HMG-CoA reductase inhibitors.

Adverse drug interaction

- ✓ rash and elevated hepatic enzymes.
- ✓ IV fluconazole...Visual disturbances (in up to 30%)
- ✓ Chronic Oral use....Photosensitivity dermatitis

Voriconazole...

Spectrum of activity.....similar to itraconazole

- Candida sp (including some fluconazole-resistant species such as *Candida krusei*)
- dimorphic fungi.
- invasive aspergillosis

Posaconazole

- ❖ Has broadest-spectrum activity
 - against most species of *Candida* and *Aspergillus*.
 - ✓ mucormycosis.
 - ✓ prophylaxis of fungal infections during
 - induction chemotherapy for leukemia, and for
 - allogeneic bone marrow transplant

Dose....dosage of 800 mg/d, divided (BID/QID)

- ❖ Absorption is increased with meals **high in fat**.
- ❖ Available with IV and SR tablet

Drug interaction.....it inhibits inhibitor CYP3A4

Isavuconazole (isavuconazonium sulfate)

- ❖ Isavuconazonium sulfate is a prodrug of isavuconazole;
- ❖ It is available as capsules and an IV formulation.
- ❖ Food does not **significantly affect absorption**
- ❖ Spectrum of activity.....similar to posaconazole.

Clinical use.... for the treatment of **invasive aspergillosis and **invasive mucormycosis**.**

Echinocandins

Drugs In this class..... Caspofungin, micafungin, and anidulafungin

MOA.....act at the level of the fungal cell wall by inhibiting the synthesis of $\beta(1-3)$ -glucan

- results in disruption of the fungal cell wall and cell death.

- ❖ active against *Candida* and *Aspergillus*, but not

- ✓ *C. neoformans* or the agents of zygomycosis and mucormycosis.

- ✓ are available only in intravenous formulations.

Caspofungin is indicated for

- ✓ disseminated and mucocutaneous candidal infections
- ✓ empiric antifungal therapy during febrile neutropenia (now replaced AMB)
- ✓ invasive aspergillosis only as salvage therapy
 - **in patients who have failed to respond to amphotericin B**

Micafungin is indicated for

- ✓ mucocutaneous candidiasis,
- ✓ candidemia, and
- ✓ prophylaxis of candidal infections in bone marrow transplant patients.

Anidulafungin.....esophageal candidiasis and invasive candidiasis, including candidemia.

Adverse Effects

❖ Echinocandin agents are extremely well tolerated

But minor gastrointestinal **side effects** and flushing

Caspofungin +cyclosporine

Elevated liver enzymes.....combination should be avoided.....decrease cyclosporine level

Micafungin has been shown to (+)

✓ levels of nifedipine, cyclosporine, and sirolimus.

Anidulafungin does not have significant drug interactions, but

—histamine release may occur during **IV** infusion.

Oral systemic antifungal drugs for mucocutaneous infections

Drugs in this class: Griseofulvin, Terbinafine

Griseofulvin

- ❖ Mechanism of Griseofulvin'sat the cellular level
 - ✓ inhibits microtubule function.....disrupts fungal cell division
- ❖ Deposited in newly forming skin binds to keratin, protecting the skin from new infection.
 - ✓ must be administered for 2–6 weeks for skin and hair
- ❖ Nail may require therapy for months
- ❖ Absorption is improved with **fatty foods**

Griseofulvin

Adverse effects include

- ✓ an allergic syndrome much like
- ✓ serum sickness,
- ✓ skin reactions,
- ✓ a lupus-like syndrome,
- ✓ Hepatotoxicity
- ✓ Drug interactions....**increase** enzymes **which metabolizes**
 - with warfarin and phenobarbital.
- ❖ Griseofulvin has been largely replaced by newer antifungal medications such as **itraconazole** and **terbinafine**

Terbinafine

- ❖ Terbinafine is a synthetic allylamine
 - ❖ oral formulation.....250 mg/d.
 - ❖ It is used in the treatment of dermatophytoses, especially onychomycosis
 - ❖ It is a keratophilic medication, but unlike griseofulvin,
 - it is fungicidal.
 - for onychomycosis >>> griseofulvin or itraconazole
- MOA**...it interferes with ergosterol biosynthesis,
- inhibits the fungal enzyme squalene epoxidase
 - leads to the accumulation of the sterol squalene (toxic to the organism).
- ❖ Adverse effects....rare...GI upset and headache and and serious hepatotoxicity

Nystatin

- ❖ It is a polyene macrolide
- ❖ It is too toxic for parenteral
- ❖ only used topically.
 - ✓ available in creams, ointments, suppositories
- ❖ has little toxicity, but
 - ✓ oral use is often limited by the unpleasant taste.
- ❖ Spectrum of activityactive against most
 - ✓ Candida sp and
- ✓ common indications include
 - oropharyngeal thrush,
 - vaginal candidiasis, and
 - intertriginous candidal infections.

Topical Azoles

- ❖ clotrimazole and miconazole are used topically
 - ✓ Vaginal pessaries.....for vulvovaginal candidiasis.
 - ✓ Oral clotrimazole troches.....for treatment of oral thrush and
- ❖ both agents are useful for dermatophytic infections
 - Tinea corporis,
 - Tinea pedis, and
 - Tinea cruris.
- ❖ shampoo forms of ketoconazole
 - ✓ For seborrheic dermatitis and pityriasis versicolor.